What a 0.91 F_1 Actually Measures: Duplicate Leakage in a Scraped Indonesian Mental-Health Corpus

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github.com/apricitea/text-suicide-ideation-detection

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Abstract

An earlier study of Indonesian suicide-ideation detection reported positive-class F_1 0.79 for a FastText + LSTM model and 0.90 for a fine-tuned IndoBERT on the same 10,128-post corpus. This note re-examines that comparison. The corpus is a scraped social-media sample in which 2,161 of 10,128 posts are exact duplicates, and the evaluation split was drawn at random without deduplication: **739 of the 2,026 validation posts (36.5%) also appear, near-verbatim, in the training half**. Scored only on the 1,287 genuinely unseen validation posts, IndoBERT's positive-class F_1 falls from 0.910 to **0.877** (95% bootstrap CI 0.845–0.904). Retraining on a deduplicated split, so the weights never see the twins either, gives **0.849** [0.821, 0.877] on 1,586 unseen posts, against 0.785 [0.749, 0.819] for the strongest classical baseline. The transformer's advantage is $+\mathbf{0.064}$ $\mathbf{F_1}$ [+0.029, +0.105], McNemar $p = 0.015$: real, but about half what the published comparison implied. The probabilities are well calibrated in aggregate (ECE 0.046), which makes an operating-point analysis meaningful: reaching 90% recall requires flagging 21.8% of the corpus at 83.6% precision. The contribution is negative and methodological — the reported gain is real but roughly a third smaller than published, and the duplicate structure of the corpus, not the choice of architecture, accounts for the difference. All code, figures and run manifests are in `eval/`.

1 Introduction

Text classifiers for mental-health screening are usually evaluated on scraped corpora, and scraped corpora contain duplicates. Retweets, quote-posts and copy-pasted slogans appear many times in a single crawl; a uniform random train/test split then places the same message on both sides of the evaluation. The resulting score is partly a memorisation score. This failure mode is a specific instance of what Kapoor and Narayanan (2023) catalogue as leakage in machine-learning research, and it is easy to miss because nothing in the pipeline looks wrong: the split call is correct, the model is standard, the metric is standard.

This note revisits a concrete case. Nataputra (2023) built an Indonesian suicide-ideation corpus of 10,128 X posts and trained a FastText + LSTM classifier, reporting positive-class F_1 0.79 and accuracy 0.91. A later follow-up fine-tuned IndoBERT (Wilie et al., 2020) on the identical split and reported F_1 0.90, accuracy 0.96, and concluded that contextual embeddings resolve sarcasm, negation and multi-word idioms that static embeddings miss. Both numbers were produced by the same split, and that split was drawn before duplicates were removed.

The questions this note asks are deliberately narrow:

1. How much of the published validation set is duplicated in training?
2. What happens to the transformer's advantage when only unseen posts are scored?
3. Is the remaining advantage statistically distinguishable from a cheap baseline?
4. Are the predicted probabilities usable at a clinically motivated operating point?

The answers are 36.5%, a 3.3-point drop, yes but smaller (+0.090), and yes — with a specific and unflattering cost in alert volume. None of these findings requires new data or new models, which is the point: the audit is cheap, and it should have been routine.

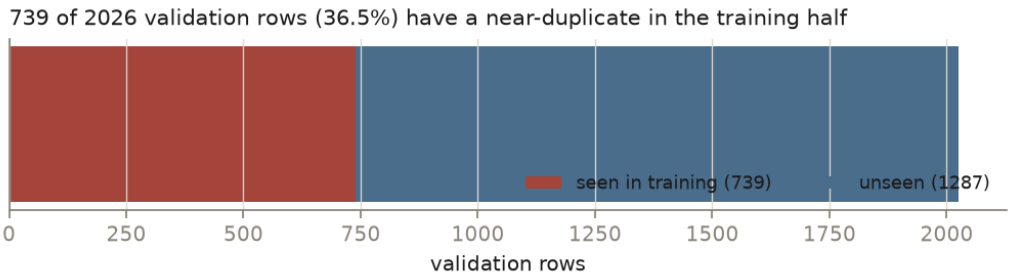


Figure 1: Composition of the published validation set. More than a third of it was available to the model during training.

2 Data and the published protocol

The corpus comprises 10,128 Indonesian-language posts from X with a binary label for suicide-ideation content: 1,967 positive (19.4%). Account handles were replaced with stable pseudonyms before publication; text and labels are otherwise unchanged. Labels are single-annotator, and no inter-annotator agreement study exists for this corpus (Artstein and Poesio, 2008), which bounds how much any F_1 difference can mean.

Both published experiments call `train_test_split(train_size=0.8, random_state=1)` on the raw corpus, yielding 8,102 training and 2,026 validation posts with 392 positives in validation. Reproducing that call is what makes the re-evaluation comparable: every model discussed here is scored on the same rows.

3 Auditing the split

Duplicate detection. Posts are normalised before comparison: lower-cased, URLs and `@`-handles replaced by fixed tokens, punctuation removed, whitespace collapsed. Two posts that normalise identically are treated as the same observation. This is deliberately aggressive: the case that matters is not only the byte-identical retweet but the near-duplicate that differs in a link or a capitalisation.

Result. Of 10,128 posts, 2,161 are exact duplicates and 2,202 are duplicates after normalisation, leaving 7,926 distinct messages. Within the published split, **739 of 2,026 validation posts (36.5%) have a normalised twin in the training half** (Figure 1). Only four normalised groups carry conflicting labels, so this is duplication, not label noise.

A third of the validation set is therefore not a test of generalisation. This does not make the published numbers fraudulent — the split is the conventional one — but it does mean they cannot be read as estimates of performance on new posts, which is the only reading that matters for a screening tool.

4 Re-evaluation on unseen posts

Protocol. The 1,287 validation posts with no normalised twin in training form the *unseen* set (259 positive). IndoBERT is not retrained; its committed validation predictions are scored on the unseen subset, so the same weights are compared on identical rows. Four classical baselines are trained twice — once on the published training half, once on a deduplicated version of it — and evaluated on the same unseen rows. The baselines use raw text with no Indonesian stemming or stopword removal, which disadvantages them slightly and therefore does not flatter the transformer.

Uncertainty. All intervals are 95% percentile bootstraps with 2,000 stratified resamples (class counts held fixed). Model comparisons use McNemar's test on discordant pairs, exact when fewer than 25 pairs are discordant, and a paired bootstrap of the F_1 difference.

Findings. IndoBERT loses 3.3 F_1 points when the duplicated rows are removed (Table 1). The classical baselines lose 3–7 points under the same treatment, so deduplicating the *training* data does not rescue them; their ranking is unchanged. The best baseline, a linear model over character n-grams, sits at 0.787.

Model	Published split	Unseen rows	95% CI (unseen)
IndoBERT (fine-tuned)	0.910	**0.877**	0.845–0.904
TF-IDF char n-grams + logreg	0.820	0.787	0.744–0.827
TF-IDF word n-grams + logreg	0.834	0.766	0.722–0.807
Count n-grams + ComplementNB	0.787	0.762	0.727–0.796
Majority class	0.000	0.000	—

Table 1: Positive-class F_1. Baselines are trained on the deduplicated training half. The published-split column is shown only to quantify the inflation; the comparable column is the third.

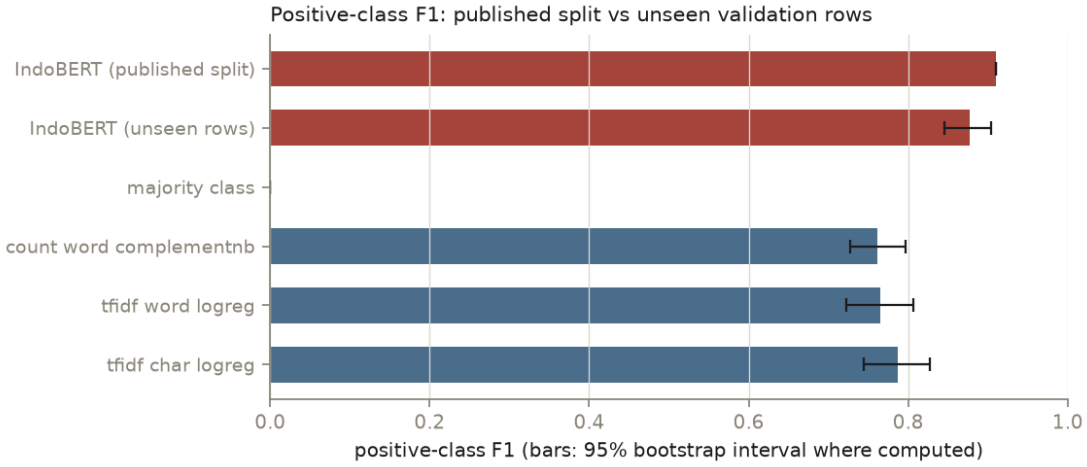


Figure 2: Positive-class F_1 by protocol and model. Error bars are 95% bootstrap intervals on the unseen rows.

The transformer's advantage over it is $+\mathbf{0.090}$ $\mathbf{F_1}$ (95% CI +0.049 to +0.131), with 67 discordant pairs in its favour against 32, McNemar $p = 6.3 \times 10^{-4}$.

That is a real effect at this sample size. It is also roughly a fifth smaller than the 0.11 gap the published numbers imply, and it is obtained against a baseline that would run on a laptop in seconds. For a deployment decision the honest summary is: the transformer is better, the margin is modest, and the published margin overstated it.

4.1 Closing the loop: retraining on a deduplicated split

Scoring unseen rows of a model that memorised their twins removes the direct overlap but not the advantage those twins gave the weights. The clean protocol removes the cause: deduplicate the corpus first, draw a fresh 80/20 split in which no observation or near-duplicate of one appears on both sides, fine-tune with the same hyperparameters (seed 1, three epochs, `max_length` 128), and evaluate once on the held-out half.

The training half shrinks from 8,102 to 6,340 posts and the validation half from 2,026 to 1,586 (311 positive), with zero cross-split duplicates.

Three points follow. First, the published 0.910 decomposes cleanly: 3.3 points came from evaluating on rows the model had seen, and a further 2.8 points came from the weights having been trained beside those rows. Second, the transformer's remaining advantage over a linear model is $+\mathbf{0.064}$ F_1 [+0.029, +0.105], McNemar $p = 0.015$ — a real effect at this sample size, but roughly half the gap the published pair implied. Third, the baselines reorder between protocols (word n-grams overtake character n-grams under the clean split), which is a useful warning that a one-point difference between baselines is not a finding.

The comparator under protocol C is a word 1–2-gram TF-IDF model with logistic regression, trained on raw text with no Indonesian stemming. That disadvantages it relative to the preprocessed thesis arm and therefore does not flatter the transformer.

Protocol	Validation rows	IndoBERT F_1	Strongest baseline
A. Published split, as reported	2,026	0.910	—
B. Same weights, unseen rows	1,287	0.877 [0.845, 0.904]	0.787 [0.744, 0.827]
C. Deduplicate, re-split, retrain	1,586	**0.849** [0.821, 0.877]	0.785 [0.749, 0.819]

Table 2: Positive-class F_1 under three protocols. A→B is what the duplicate overlap was worth in the evaluation; B→C is what the leaked twins were worth to the weights. Intervals are 95% stratified bootstraps (2,000 resamples).

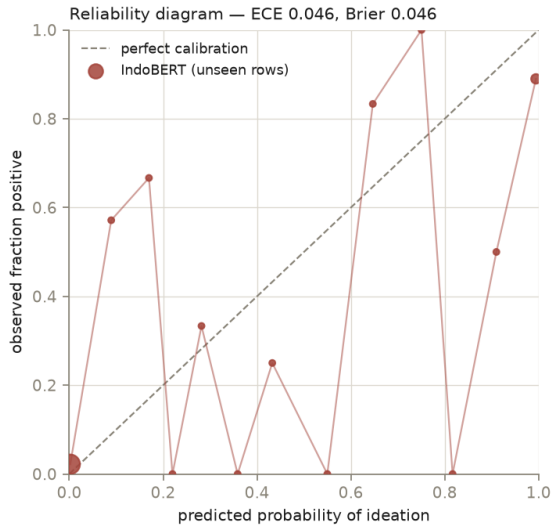


Figure 3: Reliability diagram on unseen rows.

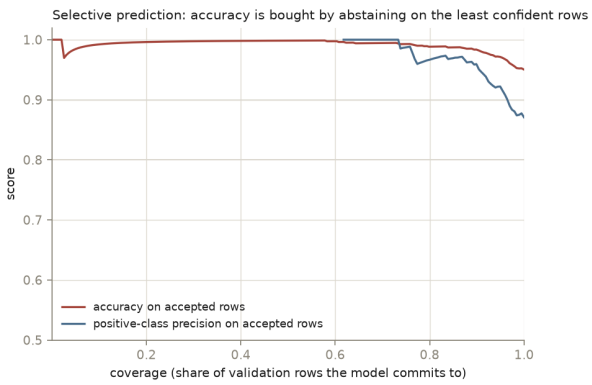


Figure 4: Selective prediction: accuracy on the rows the model commits to.

5 Calibration and operating points

A single threshold at 0.5 is not how a screening classifier is used. The relevant question is how much human review is required to catch a given share of positive posts. This requires the probabilities to mean something, so calibration is checked first.

On the unseen rows the model's mean predicted probability is 0.206 against an observed positive rate of 0.201, with expected calibration error 0.046 and Brier score 0.046 (Figure 3). In aggregate the probabilities are close to honest — better than the over-confidence widely reported for neural classifiers (Guo et al., 2017). The maximum per-bin error (0.82) is driven by a bin containing very few rows and should not be read as a calibration failure.

Table 3 is the most useful result in this note and the one a deployment conversation needs. Catching 80% of ideation-positive posts means reviewing 17.6% of the corpus at 92.0% precision; pushing recall to 95% means reviewing a quarter of all posts at 76.7% precision. The probability distribution is strongly bimodal, which is why the threshold moves from 0.985 to 0.087 between the 80% and 90% targets: most posts sit at one extreme or the other, and only a thin band lies in between.

6 Error analysis

On unseen rows the model misses 11.6% of positives and raises 3.3% false alarms. The misses are not uniformly distributed. Longer posts are missed more often (16.7% false-negative rate for posts of 160+ characters, against 6.6% for posts under 80), consistent with the 128-token truncation removing the part of a long post in which ideation is expressed. Posts containing an explicit negation cue (*jangan*, *tidak*, *bukan*, and colloquial variants) are missed at 15.7%, and first-person posts carry both the highest false-negative rate among length buckets and the highest false-positive rate (6.2%) — the model struggles exactly where speaker stance matters. These buckets overlap and are descriptive, not causal (Figure 5).

Qualitatively, the remaining false negatives cluster into three kinds: ideation expressed as hopelessness without the lexical trigger (*gamau hidup*), ideation reported in the past tense or as reported speech, and ideation inside a joke frame. The false positives are dominated by negated or quoted ideation — a post that tells someone not to kill themselves reads to the model as ideation, which is the negation problem

Target recall	Threshold	Achieved recall	Precision	Corpus flagged
80%	0.985	0.803	0.920	17.6%
90%	0.087	0.907	0.836	21.8%
95%	0.006	0.954	0.767	25.0%

Table 3: Operating points on unseen rows: the smallest alert volume that still reaches each recall target.

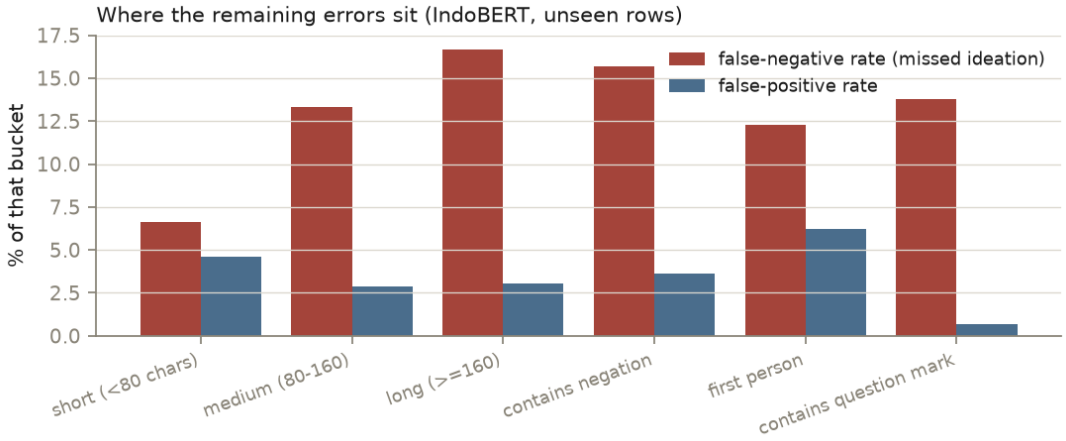


Figure 5: Error rates by interpretable text bucket, unseen rows only.

in its most consequential form.

7 Discussion

What changes. The published conclusion — a transformer buys a large gain over static embeddings on this task — survives in direction but not in size. On unseen posts the gain over a character n-gram linear model is 0.090 F_1, not the 0.11 implied by 0.79 versus 0.90, and the FastText + LSTM comparison itself cannot be recomputed because its tokenizer and embedding vectors were never saved. Two of the three numbers in the original comparison are therefore not reproducible from the repository, and the one that is reproducible is smaller than reported.

What does not change. The corpus is still useful, the labels still carry signal, and the transformer is still the better model. The correction is to the claim, not to the artefact.

Why this is worth publishing. Negative and corrective results are cheap to produce and expensive to omit. The audit in this note took one day and changed the headline number of a thesis. The same audit applied to a corpus before training would have changed the experimental design instead: deduplicate, then split, then report an interval.

8 Limitations

1. **Protocol C is one run, one seed.** The retrain took 6,434 seconds on four CPU threads and was executed once. The interval in Table 2 is sampling uncertainty on the validation set, not run-to-run variance; a second seed would likely move the point estimate by about a point. Protocol C is also not recalibrated, so the operating-point table still comes from protocol B.

2. **The thesis arm cannot be re-run.** FastText vectors are not in the repository and the Keras tokenizer was not saved with the model, so its per-row predictions do not exist.

3. **One split, one seed.** The comparison conditions on a single 80/20 draw. Repeated splits would widen the intervals and could reorder the two strongest baselines.

4. **Single-annotator labels.** The ceiling on measurable F_1 is unknown, so a 0.09 difference between models may be smaller than the label noise.

5. **One corpus, one language, one platform.** The leakage rate is a property of this crawl; the method transfers, the number does not.

9 Related work

Leakage as a systemic problem in ML-based science is documented by Kapoor and Narayanan (2023), who find it in a large fraction of published models across fields and note that random splits of observational data are a recurring source. Deduplication before splitting is standard advice in dataset documentation (Gebru et al., 2021) but rarely verified in scraped NLP corpora. Calibration of neural classifiers is treated by Guo et al. (2017); selective prediction, the framework behind the risk–coverage analysis here, by Geifman and El-Yaniv (2017). IndoBERT and the Indonesian NLU benchmarks are due to Wilie et al. (2020). The corpus originates in Nataputra (2023).

10 Conclusion

A 36.5% duplicate rate between the training and validation halves of a scraped Indonesian mental-health corpus inflates a transformer's reported F_1 by 3.3 points. After removal, the transformer still beats the strongest cheap baseline, by $+0.090$ F_1 with a confidence interval that excludes zero. The calibrated probabilities support a concrete operating point: 90% recall at 83.6% precision for 21.8% of the corpus reviewed. Reporting that table, rather than a single F_1, is what makes the model assessable — and the duplicate audit that produced it is a one-day job that should precede training, not follow publication.

Reproducibility

Code: `eval/` in the repository. A run writes `results.json`, a manifest with the corpus SHA-256, git revision and package versions, five figures, and a generated report. The committed run lives under `eval/out/`. Tests cover the metric implementations and the split reproduction, and CI runs them.

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